

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	20 July 2026
Team ID	
Project Name	Plugging Into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Technical Architecture

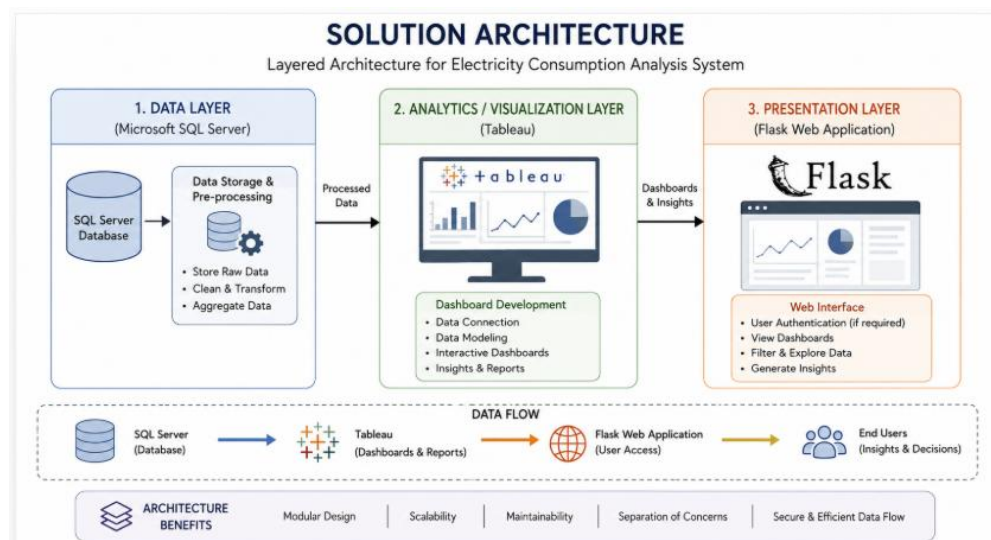


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application	HTML, CSS (served via Flask templates)
2.	Application Logic	Routing and page-serving logic	Python, Flask
3.	Database	Storage, cleaning, and querying of the dataset	Microsoft SQL Server, T-SQL
4.	Data Visualization	Dashboard and story creation	Tableau
5.	File Storage	Local storage of dataset/exports	Local filesystem
6.	External API	Not used in this project	N/A

S.No	Component	Description	Technology
7.	Machine Learning Model	Not used in this project	N/A
8.	Infrastructure	Application run locally during development	Local server (Flask dev server)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Web and visualization frameworks used	Flask, Tableau
2.	Security Implementations	Access to the database is restricted to the pipeline; no public write access	SQL Server access controls
3.	Scalable Architecture	Layered design: data, visualization, and presentation layers act independently	3-tier architecture
4.	Availability	Application runs whenever the Flask server/dashboard is active	Flask dev server
5.	Performance	Pre-aggregated queries and Tableau extracts keep dashboards responsive	SQL Server indexing, Tableau extracts